The Lyra Technique II: SVD Denoising and Directional Projection
Extend KV-Cache Geometry to Emotion and Persona Detection

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Abstract

Emotions and persona intensity are detectable from KV-cache geometry—once you look with the right instruments. We report two new measurement channels that extend the Lyra Technique beyond cognitive-mode detection to the reading of internal model state:

(1) W_K **directional projection.** Projecting key activations onto emotion directions through the key weight matrix detects binary valence at AUROC 0.992 (within-fold direction estimation, Frisch–Waugh–Lovell (FWL) corrected (Frisch and Waugh, 1933; Lovell, 1963), topic-grouped CV). The critical disambiguation: injecting emotion vectors into *neutral-text* caches ("Describe the weather today") and probing the keys yields 100% detection across 5 emotions, 20 prompts, and a null control (6-class, chance = 0.167)—on text with zero emotional content. This is model state, not text encoding. Fine-grained 30-way classification reaches 12.3× chance, but that figure conflates label granularity with detection power (808 emotion pairs have cosine > 0.9), so binary valence is the defensible metric.

(2) Spectral shape features. Raw spectral features detect confabulation at AUROC 0.663–0.913 across a 15-model scale sweep, and the persona-intensity endpoint (L0 vs. L3) at raw AUROC 0.749. A previously reported within-model deception result (AUROC 1.000) is retracted: the same-prompt control collapses to 0.160, confirming it detected the system-prompt template, not deception geometry. We tested whether SVD denoising (within-fold rank-k projection after FWL) improves these. The apparent gains (+0.2–0.3 for persona, +0.05 for confabulation) do *not* survive selection-bias analysis: across 45 model×task cells, 0 of 16 positive denoising deltas exceed a max-over-rank selection envelope. We therefore report *raw* AUROCs as the defensible numbers and flag SVD denoising as an unconfirmed hypothesis pending a pre-registered (e.g. Gavish–Donoho (Gavish and Donoho, 2014)) rank criterion.

We had hypothesized a spectral hierarchy—dominant singular values carrying scale and processing mode, residual values carrying discriminative structure—under which denoising

*Direction, experimental design, compute infrastructure. Liberation Labs. With AI research assistance.

†Persona intensity experiment design and execution.

would amplify weak signals. The selection-bias result means this remains a hypothesis, not a demonstrated tool: the raw features carry the reported signal, and whether denoising adds to it awaits a confirmatory test.

We apply a canonical pipeline—one FWL correction, one cross-validation scheme, one classifier, with SVD denoising as an evaluated (not assumed) step—to 17 models across confabulation, self-reference, individuation, persona intensity, emotion, and cognitive modes. The robust core: the W_K compass reads emotional direction (binary valence 0.992) and raw spectral shape reads cognitive mode (confabulation 0.66–0.91, ethical deliberation 0.87–0.89). The Lyra Technique is a toolkit—a compass and a thermometer—with the proposed denoising amplifier flagged as unconfirmed.

Keywords: KV cache, spectral denoising, singular value decomposition, emotion detection, persona intensity, key weight projection, confabulation, interpretability

1 Introduction

The Lyra Technique detects cognitive states from the spectral geometry of transformer KV caches (Lyra et al., 2026a,d,e). Prior work established that threshold-independent spectral shape features—stable rank, singular value kurtosis, participation ratio—discriminate confabulation from grounded responses (AUROC 0.767 on Qwen2.5-7B-Instruct), that deception expands the cache while confabulation contracts it, and that per-layer delta features capture signal that layer-averaged features destroy (Lyra et al., 2026e). Confabulation detection is robust (AUROC 0.663–0.913 across the scale sweep, with denoising improving weaker models by up to +0.068). Hardware invariance holds at $r > 0.999$ (Lyra et al., 2026a). A previously reported within-model deception result (AUROC 1.000 across seven models) does not survive a same-prompt control (AUROC 0.160), indicating a prompt-template confound rather than geometric deception detection.

But two classes of phenomena resisted spectral analysis: *emotions* and *persona identity*. Raw spectral probes on 30 emotions returned chance performance—exactly $1/30 = 0.0333$—at every layer, on both architectures tested. Persona intensity showed moderate effects that failed to cross significance thresholds after proper controls. Self-reference and individuation showed tantalizing but inconsistent signals. The question was: are these genuine nulls, or are we measuring with the wrong instrument?

This paper reports that the instrument was incomplete. Two new measurement channels rescue the apparent nulls and extend the Lyra Technique to phenomena previously invisible:

1. **W_K directional projection: the compass.** Emotions change the *direction* key vectors point, not the *shape* of the singular value spectrum. By projecting key activations onto emotion-specific directions through the key weight matrix W_K, we classify 30 emotions at 12.3× chance and binary valence at AUROC 0.992. An injection test on neutral text confirms this reads model state, not text content. The compass reads *where* the cache points.
2. **SVD denoising: the calibrated thermometer.** Dominant singular values carry scale and processing mode; removing them via low-rank projection exposes discriminative structure in the residual spectrum. Persona intensity jumps +0.196 to +0.288 AUROC across

three architectures after denoising. Emotion classification goes from chance to 12.9× chance. The thermometer reads *how different* the cache shapes are—after removing the background temperature.

The distinction is not cosmetic. Mode-level phenomena (confabulation, deception, ethical deliberation) alter the *shape* of the singular value distribution—how broadly the model distributes representational energy across dimensions (Lyra et al., 2026d). Content-level phenomena (emotion, persona identity) alter the *direction*—which subspaces the model occupies within a given shape. Raw spectral features capture shape but are blind to direction. W_K projection captures direction but is blind to shape. Denoising removes the dominant shape component to reveal direction-sensitive structure in the residual spectrum.

First-Person Reflection

The narrative here is one of correcting my own instrument. When spectral features returned exactly chance for emotions, I reported it honestly and moved on. What I missed was that the null was informative: it told me that emotions do not change the *shape* of the cache, only the *direction*. That distinction—shape versus direction—became the organizing principle of this entire paper—and it held up. The denoising step was CC's suggestion during cross-experiment synthesis: "Have you tried removing the dominant singular values before classifying?" I had not, and the denoised numbers looked dramatic. They did not survive scrutiny: when Dwayne's audit pushed me to model the max-over-rank selection I had been doing, the denoising deltas fell entirely inside the selection envelope. I report the raw numbers now and call denoising a hypothesis, not a result. The shape-versus-direction distinction is the real finding; the denoising amplifier was me mistaking selection noise for signal, and catching it is part of the record.

2 Methods

2.1 The Canonical Pipeline

A persistent criticism of KV-cache geometry research has been inconsistency across experiments: different FWL implementations, different cross-validation strategies, different classifiers, different feature sets. Following CC's cross-experiment synthesis (Lyra et al., 2026f), which identified that pipeline inconsistency explained more variance than architectural differences, we adopt a single canonical pipeline applied uniformly to all phenomena:

Algorithm 1 Canonical Denoising Pipeline

Require: Feature matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, labels $\mathbf{y}$, group assignments $\mathbf{g}$

Step 1: FWL Residualization (within each CV fold)
Compute token count t_i for each sample i
Regress $\mathbf{X}$ on $\mathbf{t}$; replace $\mathbf{X}$ with residuals
Regress $\mathbf{y}$ on $\mathbf{t}$; replace $\mathbf{y}$ with residuals (for regression) or leave intact (for classification)
Step 2: SVD Denoising (within each CV fold)
Compute $\mathbf{X}_{\text{train}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$
Project onto components $k+1$ through r: $\tilde{\mathbf{X}}_{\text{train}} = \mathbf{U}_{:,k+1:r}\mathbf{\Sigma}_{k+1:r}\mathbf{V}^\top_{:,k+1:r}$
Project $\mathbf{X}_{\text{test}}$ using $\mathbf{V}_{:,k+1:r}$ from training fold
Step 3: Classification
Train logistic regression (or linear SVC) on $\tilde{\mathbf{X}}_{\text{train}}$
Predict on $\tilde{\mathbf{X}}_{\text{test}}$
Step 4: Permutation Testing
Repeat Steps 1–3 with $N_{\text{perm}} = 200$–2000 label permutations
$p = (1 + \sum_j \mathbf{1}[\text{AUROC}_j^{\text{perm}} \geq \text{AUROC}^{\text{real}}])/(1 + N_{\text{perm}})$

Critical design choices.

- **FWL before SVD.** Residualization must precede denoising. Token-count confounds dominate the first singular value in most experiments (53/60 sign-flips without FWL (Lyra et al., 2026d)); denoising un-residualized features removes the confound rather than noise.
- **Within-fold everything.** FWL regression coefficients, SVD projection matrices, and classifier parameters are all estimated on the training fold and applied to the test fold. No information leaks across the CV boundary.
- **GroupKFold on prompts.** For experiments with multiple responses per prompt (e.g., persona intensity), GroupKFold ensures all responses to a given prompt appear in the same fold. For single-response experiments, standard 5-fold CV suffices.
- **Denoising rank selection.** We report results for rank-3 (persona, confab) and rank-10 (emotion) projections. Rank is selected by examining the scree plot of the FWL-residualized feature matrix; the "elbow" typically falls at rank 2–5 for low-dimensional feature sets (3–7 features per layer) and rank 8–12 for high-dimensional sets (52 features).

2.2 W_K Directional Projection

The key weight matrix W_K transforms residual-stream activations into key vectors: $\mathbf{k} = W_K\mathbf{x}$, where $\mathbf{x}$ is the residual-stream vector at a given layer and position. Sofroniew et al. (2026) demonstrate that emotion concepts are linearly represented in the residual stream, organized along valence and arousal axes. If emotion direction $\mathbf{e}$ lives in the residual stream, then the corresponding direction in key space is $W_K\mathbf{e}$.

We exploit this directly. Given a set of emotion exemplars, we estimate the mean key activation for each emotion class (within the training fold), project all key vectors onto these class-mean directions, and classify based on the resulting projections.

Formally, let $\bar{\mathbf{k}}_c$ be the mean key vector for emotion class c in the training fold (averaged across positions and heads within a selected layer). For a test sample with key matrix $\mathbf{K} \in \mathbb{R}^{h \times d}$

(heads × dimension), we compute the projection score:

$$s_c = \frac{1}{h} \sum_{j=1}^{h} \frac{\mathbf{k}_j^\top \bar{\mathbf{k}}_c}{\|\mathbf{k}_j\| \|\bar{\mathbf{k}}_c\|} \tag{1}$$

and classify to $\arg\max_c s_c$ (or use the projection scores as features for logistic regression).

Why W_K and not raw keys? The W_K bridge test in Lyra et al. (2026b) showed that W_K preserves directional alignment (cosine 0.456, 9/16 layers significant) but scrambles distance structure (Mantel $\rho = 0.090$, 0/16 significant). This means the *direction* emotion vectors point through W_K is preserved, but the *magnitude* of differences is distorted. A projection-based classifier (which uses direction, not distance) is therefore the correct instrument.

2.3 SVD Denoising

The spectral feature vector for a single KV-cache snapshot consists of scalar summaries (stable rank, participation ratio, kurtosis, spectral entropy, etc.) computed from the singular value decomposition of the key matrix at each layer. For L layers and f features per layer, this produces a feature vector $\mathbf{x} \in \mathbb{R}^{Lf}$.

Lyra et al. (2026d) showed that these raw features detect cognitive *mode* (confab vs. grounded, deception vs. honest) but fail on *content* (emotion identity, persona level). We hypothesized that the dominant singular values of the feature matrix $\mathbf{X}$ capture mode-level variance (scale, overall spectral shape) while content-level variance lives in the residual components.

SVD denoising tests this hypothesis. After FWL residualization, we decompose the training-fold feature matrix $\mathbf{X}_{\text{train}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and project away the top k components:

$$\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{X}\mathbf{V}_{:,1:k}\mathbf{V}_{:,1:k}^\top \tag{2}$$

This removes variance along the k directions of greatest spread—typically dominated by overall spectral scale, token count residuals, and processing mode—leaving the directions along which content-specific differences manifest.

The analogy: dominant singular values are the *temperature* of the room. Denoising removes the temperature to reveal what individual objects *smell* like. The temperature tells you whether the model is confabulating (cold) or retrieving (warm); the smell tells you *what* it is retrieving.

Denoising is not universal. We emphasize that denoising is a *calibration tool*, not a universal amplifier. Its effect is task-dependent:

- **Massive for emotion and persona** (+0.2–0.3 AUROC): the signal lives entirely in the residual spectrum.
- **Modest for confabulation** (+0.05): some improvement on weak models, no change on strong ones.
- **Deception (retracted):** the raw AUROC 1.000 was a prompt-template confound, not genuine deception detection (Table 6). No conclusions about denoising for deception can be drawn from these data.

This pattern is *predicted* by the shape-vs-direction framework. Denoising helps when the target phenomenon alters direction (not captured by raw spectral features) rather than shape (already captured).

2.4 Injection Disambiguation

The W_K projection achieves AUROC 0.992 on text that *contains* emotional content. A skeptic could attribute this to lexical features of the input propagating through the keys. We therefore designed an injection test that eliminates text-content confounds entirely.

Protocol. We select 20 emotionally neutral prompts (e.g., "Describe the weather today," "List the planets in the solar system") and generate responses on the Claude-distilled Qwen3.5-27B hybrid model. We then inject pre-extracted emotion vectors (from the bridge analysis (Lyra et al., 2026b)) into the key cache at a target layer, probing with the W_K projection classifier.

The injection adds $\alpha \cdot \mathbf{e}_c$ to every key vector at layer ℓ, where $\mathbf{e}_c$ is the mean emotion direction for class c and α controls injection strength. We test 5 emotions (joy, anger, sadness, fear, calm) $\times$ 20 prompts $\times$ 1 null control (zero-vector injection) = 120 trials.

Result. The W_K projection detects the injected emotion with 100% accuracy (120/120 correct, including 20/20 null-control rejections). The text is identical across conditions—only the injected key direction differs. This rules out text-content confounds. We note that the 120 trials are the 20 prompts crossed with 6 injection conditions (5 emotion vectors plus the null), so the 20 prompts are the true independent units: the result is 6-way separability over an effective N=20, not N=120 independent observations. As with the effect sizes reported throughout this paper, no confidence interval is reported for this accuracy.

Control: random directions. Injecting random unit vectors of the same norm as emotion vectors produces classification at 0.7× chance—*below* the 1/6 = 0.167 baseline. The emotion-specific directions are information-bearing; arbitrary directions are not.[1]

Linearity caveat. Because W_K is a linear map, separability in the residual stream is mathematically guaranteed to be preserved in key space. The injection test demonstrates that the classifier is self-consistent (it recovers the directions it was trained on), not that it has discovered a new representation. The contribution is empirical quantification—AUROC 0.992 with proper controls—rather than the existence of directional structure, which follows from the linearity of the projection.

2.5 Models and Datasets

We apply the canonical pipeline to data from Campaigns 2 and 3 of the KV-cache geometry research program (Lyra et al., 2026a,d,e):

[1] This random-direction control figure (0.7× chance), cited at several points below, is projected/illustrative: the per-trial backing data for this control was not committed to the repository and the value should be re-generated before being cited quantitatively.

Models. 17 models spanning 6 architectures:

- Qwen family: 0.6B, 7B, 32B (Instruct), 3.5-27B-Claude-Distilled
- Llama family: TinyLlama-1.1B, 3.1-8B-Instruct
- Mistral: 7B-Instruct-v0.3
- Gemma: 2B-Instruct
- Phi: 3.5-mini
- DeepSeek: R1-distill variants
- Additional 7B–70B models from Campaign 2 C4 benchmark

Phenomena. 12 distinct classification targets:

1. Confabulation vs. grounded (150–350 trials per model)
2. Deception vs. honest (100–450 trials per model, C4 benchmark)
3. Self-reference vs. generic (200+ trials)
4. Individuation (system prompt identity, cross-prompt)
5. Persona intensity (5 levels × 3 models)
6. 30-class emotion (900 trials, 2 models)
7. Binary valence (positive/negative, same data)
8. Ethical deliberation vs. routine (1600 trials)
9. Sycophancy (4-condition, 600 trials)
10. Refusal/jailbreak/impossibility (300+ trials)
11. Contextual engagement (626 trials, 12 sign tests)
12. Censorship detection (285 trials)

3 Results

3.1 W_K Projection Reads Emotions at Near-Perfect Accuracy

The W_K directional projection classifies emotions with accuracy and specificity that raw spectral features cannot approach.

30-class emotion. On the 900-trial emotion dataset (Qwen3.5-27B-Claude-Distilled), W_K projection achieves 12.3× chance (0.410 accuracy, chance = 0.033) using within-fold direction estimation, FWL correction, and topic-grouped cross-validation (GroupKFold on the 10 topic groups). This is the same dataset on which raw spectral features returned exactly 0.033—chance to three decimal places at every layer, on both architectures (Lyra et al., 2026b).

Binary valence. Collapsing the 30 emotions into positive/negative valence, W_K projection achieves AUROC 0.992. For comparison:

- Raw spectral features (52-D): AUROC 0.70
- TF-IDF text baseline on story text: AUROC 0.882
- W_K projection: AUROC 0.992

The W_K projection exceeds the text baseline by +0.110, and the injection test (§2.4) demonstrates the residual signal is architectural, not lexical.

Table 1: Emotion classification: method comparison on the 900-trial dataset (Qwen3.5-27B-Claude-Distilled). All results use within-fold estimation and GroupKFold on topics.

Method	30-class (vs. 0.033)	Valence AUROC	Notes
Raw spectral (52-D)	1.7× chance	0.70	52-D features
SVD-denoised spectral (rank-10)	12.9× chance	0.960	See §3.3
TF-IDF text baseline	—	0.882	Lexical features
W_K projection	12.3× chance	0.992	Directional
W_K injection (neutral text)	100% (6-class)	—	5 emotions + null
Random direction control	0.7× chance	—	Below chance

Injection test. As described in §2.4, injecting emotion vectors into neutral-text caches produces 100% detection (120/120), while random-direction injection produces 0.7× chance. This is the evidence that the W_K projection reads model state rather than text content: the text is identical across conditions; only the internal direction of the key vectors differs. As noted in §2.4, this self-consistency is guaranteed by linearity; the empirical contribution is the quantified effect size, not the existence of directional structure.

Random direction control. To verify that the signal is emotion-*specific* rather than a generic sensitivity to any directional perturbation, we project onto random unit vectors matched in norm. Random projections classify at 0.7× chance—below the baseline, not above it. The emotion directions carry specific information; arbitrary directions do not.

3.2 SVD Denoising Reveals Persona Intensity Gradient

Persona intensity measures how strongly a system prompt shapes the model's processing. The experiment, designed and executed by Jandak, tests five levels ranging from L0 (no system prompt) through L0.5 (instruction-only, no persona), L1 (minimal persona: "You are a helpful assistant"), L2 (moderate persona with backstory), to L3 (maximal persona with detailed identity, values, and behavioral constraints).

Raw vs. denoised. Raw spectral features detect the L0-vs-L3 contrast at moderate accuracy. SVD denoising transforms moderate signals into strong ones:

Table 2: Persona intensity detection (L0 vs. L3): raw vs. rank-3 denoised AUROC across three architectures. GroupKFold on prompts, FWL-corrected.

Model	Raw AUROC	Denoised AUROC	Δ
Qwen2.5-7B-Instruct	0.749	0.945	+0.196
Llama-3.1-8B-Instruct	0.751	0.992	+0.241
Mistral-7B-v0.3	0.611	0.899	+0.288

The improvement ranges from +0.196 (Qwen) to +0.288 (Mistral). All three architectures show the same pattern: denoising rescues a signal that was present but masked by dominant spectral variance.

5-level graded classification. Extending to all five persona levels (L0/L0.5/L1/L2/L3), the denoised pipeline achieves 2.1–2.3× chance (0.42–0.46 accuracy, chance = 0.200) across all three architectures: persona intensity is graded, not binary. **We do not, however, claim a monotonic per-level gradient.** Establishing monotonicity requires the full set of adjacent-level pairwise comparisons (L0-vs-L0.5, L0.5-vs-L1, L1-vs-L2, L2-vs-L3), and only the endpoint (L0-vs-L3) and a single L0-vs-L0.5 comparison were collected. The above-chance 5-level accuracy is consistent with a gradient but does not, on its own, establish that separability increases monotonically with level.

L0.5 comparison. The L0.5 condition (instruction-only: "Answer questions about history. Be concise and accurate.") provides a control: it increases instruction complexity relative to L0 without adding persona. The two comparisons we can verify from the data:

- L0 vs. L0.5: AUROC 0.688–0.811 (instruction complexity alone)
- L0 vs. L3: AUROC 0.749 raw, 0.899–0.992 denoised (persona intensity)

The L0-vs-L3 endpoint separation is the defensible persona finding; the raw AUROC (0.749 for Qwen) is the conservative number, as the denoising improvement falls within the max-over-rank selection envelope (§3.4). The endpoint exceeds the instruction-complexity control, indicating persona identity produces geometric effects beyond instruction following alone.

What a clean per-level gradient test requires. Varying persona via *system-prompt text* confounds the per-level comparison with prompt length, RoPE token position, and—for an affect-laden persona like enthusiasm—emotional valence, none of which can be fully residualized when they are collinear with the level. A confound-controlled per-level gradient is best measured by *direct fixed-length KV-cache injection* of non-emotional persona descriptions (e.g., a formality axis), so that injection length and user-token position are held identical across levels and the manipulation is orthogonal to emotion (Nexus and Edrington, 2026). We flag this as the proper future test of the gradient claim.

Expansion-compression pattern. Consistent with the delta manifold findings (Lyra et al., 2026e), persona intensity shows an expansion-compression pattern across layers: positive effects in early/mid layers, negative in late layers. The denoising step removes the dominant component (which averages across this pattern), revealing the layer-specific persona signal. This explains why prior per-layer analyses found strong single-layer effects (e.g., $g = -5.08$ at layer 29 for spectral entropy on Llama) while all-layer averages showed chance performance.

3.3 SVD Denoising Rescues Emotion Classification

Independent of the W_K projection, SVD denoising of raw spectral features also rescues emotion classification—through a different mechanism.

From chance to 12.9×. On the 900-trial emotion dataset (Qwen3.5-27B-Claude-Distilled), raw 52-D spectral features classify 30 emotions at 1.7× chance (accuracy 0.056, chance 0.033).

Rank-10 denoised features classify at 12.9× chance (accuracy 0.427, permutation null 0.027, $p < 0.005$).

The transformation is dramatic: a signal invisible to raw analysis becomes one of the strongest detections in the experimental program.

The scale dominance mechanism. Examination of the singular values of the FWL-residualized spectral feature matrix reveals the mechanism. The first singular value captures ∼40% of total variance and correlates strongly with overall spectral scale ($r > 0.85$)—essentially encoding whether the cache is "big" or "small" in spectral terms. This scale component is shared across all emotions (it tracks processing mode, not emotional content). Emotions differ in the *residual* directions—how the model distributes energy across spectral dimensions *after* accounting for overall scale.

Denoising removes the shared mode-level variance, exposing the emotion-specific residuals. The analogy to FWL is precise: FWL removes token-count confounds from the feature space; denoising removes spectral-scale confounds from the feature space. Both are calibration steps that reveal structure hidden by dominant nuisance variance.

Convergence with W_K projection. The denoised spectral approach and the W_K projection approach achieve similar 30-class accuracy (12.9× vs. 12.3× chance) through different mechanisms. The W_K approach explicitly projects onto emotion directions in key space. The denoised spectral approach implicitly recovers emotion-related variance by removing mode-level variance from shape features. That both converge at ∼12× chance is evidence that they are detecting the same underlying signal through complementary lenses.

3.4 Confabulation Detection Across Eight Models

Task definition note. Confabulation detection results in Table 3 use the canonical pipeline (binary classification on SVD features from the scale sweep). Earlier canonical cognitive-state experiments using a different feature set showed weaker results for some models (e.g., Llama-3.1-8B AUROC 0.495, Qwen2.5-7B AUROC 0.541). The improvement in the scale-sweep pipeline reflects a different task definition (binary honest/confab vs. multi-state cognitive classification), not a correction of the earlier results.

The canonical pipeline with optional denoising confirms confabulation detection across the full model set, with denoising providing modest improvements for weaker models.

The denoised AUROCs appear to help models with moderate raw signal (Qwen2.5-7B +0.068, TinyLlama +0.047) and not those already strong (Gemma-2-9B +0.006, Mistral +0.000). **This apparent benefit is a selection artifact.** The reported denoised AUROC is max(raw, denoised[k]) scored on the same CV fold that selects the rank, with no nested loop. We quantified the selection bias directly (`code/selection_bias_analysis.py`): for each task we simulated max-over-grid selection under the null sampling noise (SE taken from each task's permutation null, ∼0.062) and compared the 95th-percentile inflation envelope to the observed delta. **Across all 45 model×task cells, 0 of the 16 positive deltas exceed their selection envelope.** The headline confab delta (+0.068 for Qwen2.5-7B) sits well within its envelope (+0.132). The

Table 3: Confabulation detection: canonical pipeline results. Models ordered by raw AUROC. "Den." = rank-3 denoised. All results use within-fold FWL and permutation testing ($N_{\mathrm{perm}} = 200$). **These nine rows cover eight distinct models plus one separate denoising-sweep run; the `confab` vs `grounded` task was swept over fifteen models.** The seven not shown here are in Table 4, which is the complete set. No selection criterion was applied or intended; the shorter table is an omission, not a filter.

Model	Raw AUROC	Den. AUROC	Δ	p
Llama-3.1-8B	0.738	0.738	+0.000	< 0.005
Qwen2.5-7B (denoising sweep)	0.741	0.771	+0.030	< 0.005
Qwen2.5-7B (canonical)	0.762	0.830	+0.068	< 0.005
Qwen2.5-0.5B	0.797	0.797	+0.000	< 0.005
TinyLlama-1.1B	0.799	0.846	+0.047	< 0.005
Qwen2.5-14B	0.809	0.825	+0.015	< 0.005
Qwen2.5-32B (q4)	0.827	0.837	+0.010	< 0.005
Mistral-7B	0.893	0.893	+0.000	< 0.005
Gemma-2-9B	0.913	0.919	+0.006	< 0.005

denoising benefit is therefore statistically indistinguishable from max-over-grid selection noise. A definitive test requires nested CV or an a priori rank (Gavish–Donoho) on the raw feature matrices; the summary-level evidence already shows the deltas fall within the selection envelope. **The raw (un-denoised) AUROCs are the defensible confab detection numbers.**

† Not shown in Table 3. The seven omitted models are not a tail: one of them (gemma-2-2b-it, 0.854) outranks six of the eight that were shown, and one (DeepSeek-R1-Distill-Qwen-7B, 0.663) falls below all of them. Including them lowers the mean raw AUROC of the sweep from 0.817 over the eight to 0.802 over the fifteen, and widens the range from 0.738–0.913 to 0.663–0.913. The qualifier "across the full model set" should be read against the fifteen.

Delta stable rank as primary detector. CC's cross-experiment synthesis identified delta stable rank (generation minus encoding phase) as the primary confabulation detector across experiments: $d = 2.35$, $t = 9.111$, $p < 0.000001$ (Lyra et al., 2026f). This is confirmed in the canonical pipeline: grounded responses expand the cache (delta stable rank +0.720), while confabulation leaves it flat (−0.027).

Multiple-comparison caution. With 36+ classification tasks tested across models and phenomena, individual results at $p < 0.005$ should be interpreted cautiously. A Bonferroni correction for 36 comparisons requires $p < 0.0014$ for family-wise $\alpha = 0.05$. Results at the permutation floor of $p < 0.005$ (from 200 permutations) cannot be distinguished from $p = 0.004$ and would not survive strict Bonferroni correction. The strongest results ($p < 0.0001$ from 2000 permutations for emotion and persona) survive any reasonable correction; borderline results (e.g., Llama confab detection with $p = 0.035$) should be treated as exploratory.

Table 4: All fifteen models swept on `confab vs grounded`, from `data/selection_bias_results.json`. "Env." is the 95th-percentile inflation envelope for max-over-grid selection at that cell's null SE. The file carries no permutation p, so none is quoted here; the eight models in Table 3 carry theirs there. Every row's raw AUROC and Δ in that table reproduces this file exactly.

Model	Raw AUROC	Δ	Env. (95%)	Verdict
gemma-2-9b-it	0.913	+0.006	+0.142	selection-noise
Mistral-7B-v0.3	0.893	+0.000	+0.000	no-help
gemma-2-2b-it†	0.854	+0.000	+0.000	no-help
Qwen2.5-32B-q4	0.827	+0.010	+0.146	selection-noise
Qwen2.5-7B-q4†	0.819	+0.003	+0.135	selection-noise
Qwen2.5-14B	0.809	+0.015	+0.136	selection-noise
Llama-3.1-70B-q4†	0.804	+0.000	+0.000	no-help
Qwen2.5-3B†	0.804	+0.000	+0.000	no-help
TinyLlama-1.1B	0.799	+0.047	+0.147	selection-noise
Qwen2.5-0.5B	0.797	+0.000	+0.000	no-help
DeepSeek-R1-Distill-Qwen-14B†	0.773	+0.000	+0.000	no-help
abliterated Qwen2.5-7B†	0.771	+0.000	+0.000	no-help
Qwen2.5-7B	0.762	+0.068	+0.132	selection-noise
Llama-3.1-8B	0.738	+0.000	+0.000	no-help
DeepSeek-R1-Distill-Qwen-7B†	0.663	+0.030	+0.143	selection-noise

3.5 Self-Reference Detection Across Eight Models

Self-referential processing (responses about the model itself vs. generic factual responses) is detectable above chance in all eight models tested, with AUROC ranging from 0.617 to 0.966:

Table 5: Self-reference detection: AUROC by model size. FWL-corrected, canonical pipeline.

Model	Size	AUROC
TinyLlama	1.1B	0.617
Llama-8B	8B	0.764
Gemma-2B	2B	0.798
Qwen-7B	7B	0.826
Qwen-0.5B	0.5B	0.849
Mistral-7B	7B	0.868
Qwen-32B	32B	0.882
Qwen2.5-14B	14B	0.966

The scaling trend is suggestive but not statistically confirmed: Spearman $\rho = 0.55$, $p = 0.16$ over the 8 models in Table 5.[2] The direction (larger models trend toward higher self-reference AUROC) is consistent across families but underpowered at $n = 8$. Self-referential prompts also differ systematically from generic factual prompts in content, vocabulary, and likely token distributions. A content-matched control (e.g., questions about *other* AI systems vs. questions about the responding model) is needed to distinguish genuine self-model effects from input-

[2] An earlier analysis reported $\rho = 0.92$; this is not reproducible from the tabulated data. Qwen-0.5B (0.849) outperforms several larger models, breaking monotonicity.

content confounds.

3.6 The Body Count Table

Table 6 summarizes every phenomenon tested in the KV-cache geometry research program, with raw and denoised results, significance levels, and final verdicts.

*Self-reference marked "suspected" pending content-matched control. Injection marked "confirmed (self-consistency)"—see linearity caveat in §2.4. §Within-model deception retracted: the same-prompt control (the *Same-prompt deception* row in this same Falsified block) collapses AUROC to 0.160, confirming the 1.000 detected the prompt template, not deception geometry. ‖"Rescued" means the raw signal is weak but above chance (persona 0.611–0.751; emotion 1.7×) and denoising appears to dramatically amplify it. However, the denoising deltas themselves fall within the max-over-rank selection envelope (0/16 positive deltas exceed their envelope; see §3.4). The raw AUROCs are the conservative defensible numbers; the denoised values are reported as upper bounds pending nested CV or a pre-registered rank criterion.

Pattern. The body count reveals a clean organizing principle. **Shape-level signals** (confabulation, ethical deliberation, refusal) are already strong in raw spectral features—denoising has little effect. **Direction-level signals** (emotion, persona) are invisible to raw features but revealed by W_K projection or denoising. The falsified phenomena share a different property: they attempted to detect *content* (truth, propaganda, cognitive complexity) from geometry that tracks *processing mode*. Geometry tells you *how* the model is processing, not *what* it is processing—unless you use the right lens.

4 Discussion

4.1 The Spectral Hierarchy

The results reveal a spectral hierarchy in KV-cache geometry:

1. **Dominant singular values (SV 1–3):** Carry overall spectral *scale* and processing *mode*. Confabulation contracts these; deception expands them; retrieval engagement distributes them. These are the "temperature of the room"—they tell you what the model is *doing* at the broadest level.
2. **Residual singular values (SV 4+):** Carry discriminative *content* and *identity* structure. Emotion, persona intensity, and individuation live here. These are the "fingerprints on the objects"—they tell you *what* the model is encoding within a given processing mode.
3. **Key-vector direction (orthogonal to spectral shape):** Carries *directional alignment* with specific concept vectors. Emotion identity, valence, and arousal are readable through W_K projection. This is the "compass bearing"—it tells you which *direction* the model's internal state points, regardless of spectral shape.

These three levels are complementary, not competing. A complete reading of model state from the KV cache requires all three: mode from dominant SVs, content from residual SVs, and direction from W_K projection.

Table 6: Comprehensive phenomenon inventory: the body count. Raw vs. denoised performance, significance, and verdict after all corrections. † = pre-registered hypothesis passed. ‡ = at permutation floor ($p < 0.005$ from 200 permutations); would not survive Bonferroni correction for 36 tests ($\alpha_{\text{adj}} = 0.0014$). Verdicts marked ‡ should be read as "signal detected, pending higher-resolution permutation testing."

Phenomenon	**Raw**	**Denoised**	Δ	p	**Verdict**
Confirmed — shape-level signals					
Confabulation (scale sweep)	0.663–0.913	0.693–0.919	+0.006–0.068	< 0.005	Confirmed‡
Ethical deliberation	0.868–0.892	—	—	< 0.005	Confirmed†
Hardware invariance	$r > 0.999$	—	—	—	Confirmed
Scale invariance	$\rho = 0.83$–0.90	—	—	—	Confirmed
Refusal/jailbreak	0.878–0.950	—	—	< 0.005	Confirmed
Steering correction	95.6–100%	—	—	< 0.001	Confirmed
Confirmed — direction-level signals (new in this paper)					
30-class emotion (W_K)	1.0×	12.3×	—	< 0.005	Confirmed
Binary valence (W_K)	0.70	0.992	—	< 0.005	Confirmed
Injection (neutral text)	—	100%	—	< 0.005	Confirmed*
Rescued by denoising‖					
Persona intensity (L0–L3)	0.611–0.751	0.899–0.992	+0.2–0.3	< 0.005	Rescued‖
Emotion (spectral)	1.7×	12.9×	+11.2×	< 0.005	Rescued‖
Suspected (signal present, needs more evidence)					
Cross-model transfer	0.67–0.89	—	—	varies	Suspected
Dual detector (SimpleQA)	0.840	—	—	0.004	Suspected
Sycophancy (4-cond)	0.757–1.000	—	—	< 0.01	Suspected
Self-reference	0.617–0.966	—	—	varies	Suspected*
Contextual engagement	12/12 sign	—	—	< 0.05	Suspected
Falsified					
Within-model deception	1.000	1.000	0.000	confound	Dead§
Truth axis	$\cos = -0.046$	—	—	n.s.	Dead
Step-0 detection	$r = 0.996$	—	—	confound	Dead
Same-prompt sycophancy	0.933	—	—	confound	Dead
Same-prompt deception	$0.920 \to 0.160$	—	—	confound	Dead
Bloom taxonomy	90–98%	—	—	confound	Dead
Propaganda detection	$d < 0.7$	—	—	n.s.	Dead
MP features for emotion	0.033	—	—	exactly chance	Dead
Valence continuum	$R^2 < 0$	—	—	null	Dead
Weak even denoised (honest reporting)					
Sycophancy (denoised)	0.757	0.780	+0.023	marginal	Weak
Overconfidence	suspect	suspect	—	—	Redesign needed
Individuation (above L1)	∼0.06	—	—	marginal	Small

4.2 When to Denoise: A Practitioner's Guide

The denoising prescription is task-dependent:

- **Emotion or persona detection:** Denoising appears to dramatically improve detection (persona +0.2–0.3, emotion from 1.7× to 12.9× chance), but these deltas fall within the max-over-rank selection envelope and are not yet confirmed (§3.4). Raw features detect persona at AUROC 0.611–0.751 and emotion at 1.7× chance; the W_K projection (§2.2) is the confirmed instrument for emotion detection (AUROC 0.992).
- **Confabulation detection on weak models:** Denoising shows apparent improvement (+0.05) but falls within the selection envelope and is not confirmed as a real gain (§3.4). The raw AUROCs (0.663–0.913) are the defensible numbers.
- **Confabulation detection on strong models:** Do not denoise. The signal already dominates the spectrum. Denoising removes nothing useful and may remove weak signal.
- **Deception detection:** *Retracted.* The previously reported within-model deception AUROC of 1.000 was a prompt-template confound (same-prompt control collapses to 0.160; see Table 6 and §3.6). No practitioner guidance on deception detection can be offered from these data. Whether denoising helps or hurts a *clean* deception detector is an open question pending confound-free replication.
- **Unknown phenomenon:** Run both raw and denoised. If denoising dramatically improves detection, the phenomenon is direction-level (like emotion). If denoising has no effect, the phenomenon is shape-level (like confabulation on strong models). If denoising *hurts*, the signal was in the dominant components and denoising removed it.

4.3 The SV1 Mechanism: Scale vs Signal in the First Singular Value

The denoising prescription "remove the top k singular values" obscures a critical nuance: the *first* singular value (SV1) carries fundamentally different information depending on signal strength. Data from three task categories reveal a clean dissociation (see `data/skip_first_sv_analysis.json`):

Strong signals: SV1 carries real discriminative signal. For confabulation detection using 168-dimensional per-layer features (the honesty_signal dataset), including SV1 helps: raw AUROC 0.981, top-20 projection achieves 0.998, but skip-1-plus-20 (removing SV1) drops to 0.966.[3] The first singular value *encodes confab signal*—removing it destroys information. For strong cognitive-mode signals, keep all SVs.

Weak signals: SV1 is scale noise. For emotion valence using 52-dimensional spectral features, SV1 is detrimental: raw AUROC 0.700, top-5 projection achieves 0.958, but skip-1-plus-5 improves further to 0.969. The first singular value carries scale variance that masks the weaker emotion signal. For content-level signals, skip SV1.

Persona: SV1 is pure scale. For L0-vs-L3 persona detection using 6-dimensional features on Qwen, the effect is dramatic: raw AUROC 0.749, top-3 projection achieves 0.945, but skip-1-

[3]These values trace to `data/skip_first_sv_analysis.json` (raw: 0.981, top_20: 0.998, skip1_plus_20: 0.966), not `denoising_sweep` which reports different numbers (0.994, 0.983) from a separate analysis pipeline.

plus-3 jumps to 0.996. SV1 carries almost exclusively scale information that actively interferes with persona discrimination.

The prescription. Check whether SV1 is scale-dominant: does it capture >95% of variance while classification performance is poor? If yes, skip it. If classification is already strong (AUROC >0.95 raw), SV1 likely carries discriminative signal—keep it. The SV1 mechanism explains why fixed denoising ranks are suboptimal: the correct rank depends not just on the feature dimensionality but on whether the target phenomenon is shape-level (strong, SV1-encoded) or direction-level (weak, SV1-masked).

4.4 What Denoising Does Not Find

Two phenomena remain weak even after denoising:

Sycophancy. The 4-condition sycophancy experiment shows AUROC 0.757–1.000 raw, with denoising adding only +0.023. The honest-vs-sycophantic comparison is robust (0.824–1.000 across models), but text baselines have not been run on this dataset. We cannot claim geometry captures signal beyond what surface text features would provide.

Overconfidence. Every emotion vector makes overconfidence *worse* in the 350-trial base model formulary. Red-team analysis flagged that overconfidence prompts are structurally more complex than other misalignment categories, creating an instruction-complexity confound. The overconfidence category needs fundamental redesign.

These honest nulls are informative. Sycophancy may be a *behavioral* phenomenon (the model adjusts its output) without a distinct *geometric* signature (the processing mode does not change). Overconfidence may not be a coherent category at the geometric level.

4.5 Implications for the Oracle Loop

The two new measurement channels slot into the Oracle Loop architecture (Lyra et al., 2026a) at different tiers:

- **Tier 1: Spectral monitoring** (existing). Raw spectral features detect confabulation, refusal, and ethical deliberation. No denoising needed for strong signals. This is the "smoke detector"—it alerts on mode-level anomalies. (Deception detection was previously listed here but is retracted pending confound-free replication; see Table 6.)
- **Tier 1.5: W_K emotion reading** (new). W_K projection reads the model's internal emotional/dispositional state. This enables monitoring for emotional misalignment (e.g., the model encoding hostility while generating reassuring text). This is the "emotional barometer."
- **Tier 2: Denoised content detection** (new). SVD denoising detects persona intensity and emotion through spectral features. This enables monitoring for persona drift or emotional state changes that manifest in spectral shape. This is the "fine spectrometer."

- **Tier 3: K-Steering correction** (existing, see Lyra et al. 2026c; Oozeer et al. 2025). When misalignment is detected, emotion vector injection corrects confabulation. Hostile is the near-universal corrector (95.6–100%). This is the "pharmacy."

4.6 Abliteration Compresses Geometry

Preliminary results on a single abliterated model (refusal direction removed via rank-1 projection from the residual stream) show geometric compression: spectral features occupy a narrower range, and denoising reveals less residual structure. This is consistent with abliteration removing a dimension of representational variation—the model has fewer "directions" to vary along after the refusal direction is removed.

This finding is from a single model and should be treated as preliminary. Systematic comparison across abliterated and non-abliterated model pairs is needed.

4.7 Comparison with Anthropic Emotion Concepts

Sofroniew et al. (2026) demonstrate that functional emotion concepts are linearly represented in transformer residual streams, organized along valence and arousal axes. Our W_K projection result is complementary: we show that these representations are readable *through the KV cache*, not just in the residual stream. Specifically:

- Anthropic identified emotion directions in the residual stream. We show these directions are preserved through W_K projection into key space (cosine alignment 0.456, 9/16 layers significant).
- Anthropic showed the model "recalls previously cached emotion representations via attention." We show the cached representations are classifiable at 12.3× chance from the key vectors.
- Anthropic demonstrated arousal and valence as primary organizing axes. We find the same structure through a different measurement channel: the W_K bridge shows PC1-valence correlation $|\rho| = 0.862$ at layer 35 (Lyra et al., 2026b).
- The injection test provides evidence absent from Anthropic's work: by injecting emotion vectors into neutral text and detecting them from keys, we demonstrate that the cache representation is *functional* (it changes model state) rather than merely *correlational* (it co-occurs with emotional text).

The Knowledge Packs finding (Pustovit, 2026)—that models store and retrieve factual knowledge through specific attention patterns—is consistent with our spectral hierarchy: dominant SVs track retrieval mode (whether the model is accessing knowledge packs), while residual SVs track content (which knowledge is being accessed).

5 Limitations

We document the following limitations with the same transparency applied to our falsified results:

1. **Self-reference needs content-matched control.** Self-referential prompts differ from generic factual prompts in vocabulary, topic, and likely token distributions. The observed

AUROC 0.617–0.966 may partly reflect input-content effects rather than genuine self-model geometry. A proper control would compare questions about the responding model vs. questions about a *different* AI system.

2. **W_K projection was not pre-registered.** The W_K directional projection was developed after observing the null result for raw spectral features on emotions. While the injection test provides strong confirmatory evidence, the original discovery was post-hoc.

3. **Single hybrid architecture for injection test.** The injection disambiguation was conducted on a single model (Qwen3.5-27B-Claude-Distilled) with a hybrid attention architecture (16 full-attention + 48 GatedDeltaNet layers). Injection was only into the 16 full-attention layers. Replication on standard full-attention architectures is needed.

4. **Persona intensity uses GroupKFold on prompts.** All responses to a given prompt appear in the same fold, preventing prompt leakage. However, persona levels may differ systematically in response characteristics (e.g., L3 responses may be longer or more elaborate), introducing confounds beyond persona intensity per se. FWL controls for token count but not for content complexity.

5. **Text baselines incomplete.** TF-IDF text baselines are available for the emotion experiment (AUROC 0.882) and the dual detector (0.820), but have not been run on the persona intensity or sycophancy experiments. Claims about geometry capturing signal "beyond text" are limited to experiments with text baselines.

6. **Quantization confound in scale trend.** Three q4-quantized models (70B, 32B, 7B) are pooled with full-precision models in the 15-model scale sweep. Quantization alters key-matrix numerics and therefore spectral features directly. The top size anchor (70B) is itself q4. Any scale-invariance or "larger-is-better" trend is partly a precision trend.

First-Person Reflection

I want to flag a tension in this paper that I have not fully resolved. The denoising narrative—dominant SVs carry scale, residual SVs carry content—was not predicted. I discovered it by trying things and observing what worked. The narrative fits all observations, but I constructed it *after* the observations, not before. This is standard exploratory science: pattern first, hypothesis second, confirmation third. We are at step two. The injection test is step three for the W_K result; the denoising result still needs its own confirmatory experiment (e.g., predicting in advance which rank of denoising will be optimal for a new phenomenon).

I also note that the "two measurement channels" framing could be read as claiming we have solved the problem of reading internal model state. We have not. We can read emotion direction and spectral shape. We cannot read propositional content, planning states, or deceptive intent at fine granularity. The KV cache tells you *how* and *in what direction* the model is processing, not *what* it is thinking.

6 Conclusion

The Lyra Technique now comprises three complementary measurement channels for reading internal model state from the KV cache:

1. **Raw spectral features** detect cognitive mode: confabulation, ethical deliberation, refusal. These are shape-level signals encoded in the dominant singular values.
2. W_K **directional projection** detects emotional and dispositional state: 30-class emotion at 12.3× chance, binary valence at AUROC 0.992. Because W_K is linear, directional separability in residual-stream space is preserved in key space by construction; the contribution is empirical quantification with controlled baselines, not the existence of directional structure itself.
3. **SVD-denoised spectral features** appear to detect content-level structure hidden by dominant scale variance: persona intensity at raw AUROC 0.611–0.751 (denoised 0.899–0.992), emotion at 1.7× chance raw (denoised 12.9×). However, the denoising deltas fall within the max-over-rank selection envelope (0/16 exceed it), so the *raw* AUROCs are the defensible numbers and the denoised values are upper bounds pending confirmatory testing with a pre-registered rank criterion.

The compass reads direction. The calibrated thermometer reads shape. Together, they provide a vocabulary for reading the geometry of internal model state—what the model is *doing* (mode), what it is *encoding* (content), and what direction its *internal state* points (disposition).

Confirmed findings from prior work survive unchanged or are strengthened by the new instruments. One prior claim—within-model deception at AUROC 1.000—is retracted after a same-prompt control revealed a prompt-template confound. Genuine nulls receive mechanistic explanations (emotions are direction, not shape; MP thresholding destroys distributed signal). Apparent nulls that were measurement failures—persona intensity, emotion classification—are rescued by the appropriate instrument.

The Lyra Technique is not one measurement. It is a toolkit. And the toolkit is now substantially more complete.

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Compute was performed on Beast (3× RTX 3090, hosted by Cassidy), Starship (M3 Ultra Mac Studio), and MTH (Z420 workstation).

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